

Do Phonetic-Feature Auxiliaries Improve Brain-to-Text Decoding?

A Data Science + Neuroscience Multi-Task Study

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Intracortical brain–computer interfaces (BCIs) offer a promising route to restore communication for people who cannot speak, but accurately decoding speech from neural activity remains a challenging problem at the intersection of data science and neuroscience. This project investigates whether adding phonetic-feature supervision as an auxiliary task to a recurrent neural network trained with Connectionist Temporal Classification (CTC) can improve brain-to-text decoding on the Brain-to-Text 2025 intracortical dataset. Motivated by evidence that speech motor and sensory cortices encode articulatory structure, I augment the official GRU+CTC baseline with an additional head that predicts multi-label phonetic features (voicing, place, manner for consonants; height, backness, rounding for vowels) at the utterance level. The main branch retains phoneme-level CTC decoding with a 1-gram language model. Experiments on the T15 copy-task dataset show that the auxiliary objective yields modest but consistent gains: phoneme error rate (PER) decreases from 0.159 to 0.157 and word error rate (WER) from 0.65 to 0.63 when using a small auxiliary weight. Linear probes further reveal that phonetic features are more linearly decodable from neural activity than letters or phonemes, supporting the neuroscientific hypothesis that articulatory attributes align naturally with intracortical representations. At the same time, overall gains are small, highlighting limitations imposed by dataset scale, label noise, and model capacity. The project contributes an end-to-end reproducible pipeline, empirical evidence on phonetic-feature auxiliaries, and reflections on how data science methods can probe neural speech representations in BCIs.

Additional Key Words and Phrases: brain-to-text, intracortical neural decoding, neuroscience, phonetic features, CTC, GRU, brain–computer interface

1 Introduction

Restoring rich, fast communication for individuals who have lost the ability to speak is a central goal of modern brain–computer interface (BCI) research. Recent work has shown that intracortical Utah arrays implanted in motor and premotor cortex can support high-accuracy decoding of attempted speech, enabling participants with paralysis to generate text at conversational rates [3, 11]. At the same time, decoding neural activity directly into usable text is a complex data science problem: the signals are high-dimensional, noisy, and non-stationary across days, and the mapping from neural population activity to linguistic units is structured and context-dependent.

This project lies at the intersection of data science and neuroscience. From the neuroscience side, prior studies suggest that neural populations in speech motor cortex encode not only specific phonemes but also underlying articulatory and phonetic features, such as voicing or place of articulation [1, 5, 6]. From the data science side, modern brain-to-text pipelines combine temporal sequence models (RNNs, CNNs, transformers) with CTC loss and language model rescoring [5, 7, 9]. A natural interdisciplinary question is whether we can turn neuroscientific hypotheses about phonological structure into concrete modeling improvements.

The Brain-to-Text 2025 Kaggle competition [2] provides an ideal testbed. It releases intracortical recordings and transcripts from a single participant performing copy-speech tasks, along with a GRU+CTC baseline and a 1-gram language model. Building on my earlier milestones, where I cleaned and explored the dataset, reproduced the baseline, and ran diagnostic probes, this final report presents a multi-task CTC system that adds a lightweight phonetic-feature auxiliary head to the official baseline.

Interdisciplinary context (X = Neuroscience) The core hypothesis is neuroscientific: if intracortical neural activity is more directly aligned with articulatory features than with arbitrary orthographic labels, then encouraging the encoder to represent these features should regularize learning and slightly reduce decoding error. The project thus uses data science techniques—probing, multi-task learning, error analysis—to test a claim about how speech representations are organized in the brain and how that organization can be leveraged for BCI design.

Research questions

- (1) **RQ1:** Do phonetic-feature auxiliaries measurably improve phoneme- and word-level decoding accuracy compared to a GRU+CTC baseline?
- (2) **RQ2:** Are phonetic features more linearly decodable from intracortical activity than letters or phonemes, as measured by simple probes?
- (3) **RQ3:** What do the empirical results reveal about the structure of neural speech representations and the limitations of current intracortical BCIs?

The rest of the paper is organized as follows. Section 2 reviews related work in neural speech decoding and phonetic-feature modeling. Section 3 describes the Brain-to-Text 2025 dataset, preprocessing, model architecture, and evaluation protocol. Section 4 presents quantitative and qualitative results. Section 5 discusses neuroscientific implications, limitations, and ethical considerations. Section 6 concludes with directions for future work and broader interdisciplinary insights.

2 Related Work

2.1 Intracortical speech BCIs

Early brain-to-text systems often used electrocorticography (ECoG) to decode phonemes or words from cortical surface recordings. Herff et al. introduced a “Brain-to-Text” framework that aligned ECoG activity with phoneme sequences and used CTC-based models to produce text [5]. Anumanchipalli et al. showed that cortical activity could be mapped to articulatory kinematics and then synthesized into intelligible audio, suggesting that articulatory spaces provide a meaningful intermediate representation [1].

More recently, intracortical arrays have pushed performance further. High-resolution Utah recordings enable fine-grained decoding of attempted speech in participants who cannot vocalize [3, 4, 11]. These systems typically train deep sequence models on session-specific data and rely on careful calibration to manage non-stationarity across days.

The Brain-to-Text 2025 challenge formalizes this setting by providing a standardized intracortical dataset and baseline code [2]. The winning system for a previous Brain-to-Text benchmark introduced DCoND, a context-aware neural representation that decodes diphones and then marginalizes to phonemes before language-model decoding [7]. This work highlights the importance of modeling local phonetic context.

2.2 Temporal sequence modeling and language priors

Across both invasive and noninvasive BCIs, modern decoding pipelines often follow a three-stage pattern: neural feature extraction, temporal sequence decoding, and linguistic reconstruction [8, 9]. For temporal decoding, CTC remains a common choice to align asynchronous neural frames with phoneme or character sequences without explicit frame-level labels [5, 9]. However, the conditional independence assumptions in CTC limit its ability to capture coarticulation and long-range dependencies [7].

To compensate, recent systems increasingly adopt transformer-based encoders and incorporate strong language priors, such as n-gram or large language models (LLMs), to constrain noisy neural predictions into semantically and syntactically plausible text [7, 8, 10]. Li et al. showed that combining context-aware diphone modeling with LLM refinement significantly improves performance on an intracortical brain-to-text benchmark [7].

2.3 Phonetic and articulatory representations

A consistent theme in the neuroscience of speech is that phonetic and articulatory features are central organizing principles in cortical representations. ECoG and intracortical studies have found that neural activity patterns reflect properties such as voicing, place of articulation, and manner rather than only discrete phoneme identities [1, 5, 6]. Duraivel et al. showed that higher-resolution recordings further sharpen these feature representations and can improve decoding accuracy [4].

At the algorithmic level, many systems implicitly leverage phonetic structure through phone-based labeling, articulatory embeddings, or diphone models [1, 7]. However, explicit phonetic-feature supervision remains relatively under-explored, especially for intracortical data. My project directly targets this gap by constructing multi-hot phonetic-feature labels and using them as an auxiliary objective in a GRU+CTC pipeline. The goal is not to design a new state-of-the-art system but to empirically test whether this neuroscientifically motivated auxiliary task provides measurable regularization benefits.

3 Data and Methodology

3.1 Data Processing

3.1.1 Dataset: Brain-to-Text 2025 intracortical recordings. All experiments use the T15 intracortical copy-task dataset distributed with the Brain-to-Text 2025 Kaggle competition [2]. Recordings were collected via a 96-channel Utah array in dorsal motor cortex from a participant with paralysis, who attempted to speak cue sentences presented on each trial. For each trial, the dataset provides:

- (1) Multichannel neural time-series features (binned spike rates and related statistics).
- (2) A textual transcript of the intended sentence.
- (3) Metadata such as corpus source, recording day, and block index.

An earlier milestone focused on dataset inspection and exploratory analysis: visualizing raw neural traces, examining distributions of sentence length and speech duration, and analyzing corpus and day-level effects. Word-match ratios and accuracy metrics from the official Kaggle baseline suggested that many sentences are decoded nearly perfectly, but performance varies across corpora and recording days.

For modeling, I follow the official train/validation trial splits provided in the baseline repository. The final experiments use these splits consistently for both baseline and auxiliary models.

3.1.2 Preprocessing and label construction. Neural features are standardized following the baseline implementation. Each trial is represented as a variable-length sequence of neural frames, with per-channel normalization and day-specific affine transformations to reduce across-session variability. A temporal convolutional front-end further smooths and compresses the input sequence before feeding it into the GRU encoder.

For phoneme supervision, I apply a grapheme-to-phoneme (G2P) converter based on CMUdict to transform each transcript into a phoneme sequence. These phoneme sequences serve as targets for the CTC loss.

To construct phonetic-feature labels, I define a `phones_to_feats` mapping that assigns a multi-hot feature vector to each phoneme:

- (1) **Consonants**: voicing (voiced/unvoiced), manner (stop, fricative, nasal, approximant, affricate), place (bilabial, labiodental, dental, alveolar, postalveolar, palatal, velar, glottal).
- (2) **Vowels**: height (high, mid, low), backness (front, central, back), rounding (rounded, unrounded).

For each utterance, the union (or frequency-weighted average) of its phoneme features forms a multi-hot vector describing the articulatory composition of the sentence. These vectors are stored alongside trial identifiers in JSONL/CSV format and reused by both the auxiliary head and diagnostic probes.

3.2 Methodology

3.2.1 Baseline GRU+CTC model. The baseline model follows the official competition architecture. After preprocessing, each trial produces a sequence of neural embeddings that feed into a five-layer bidirectional GRU encoder. Let x denote the neural sequence and h_t the hidden state at time step t . A linear layer projects h_t to logits over the phoneme vocabulary, and the CTC loss is defined as

$$\mathcal{L}_{\text{CTC}} = -\log p(y \mid x),$$

where y is the target phoneme sequence. At inference time, beam-search decoding with a 1-gram language model over phonemes produces the final phoneme sequence, which is then mapped to words to compute WER.

This baseline was reproduced and validated in Milestone 2, where the phoneme error rate (PER) without language-model rescoring was approximately 0.10 (PER = 0.1019) on the validation split. This serves as a reference point for assessing the impact of the auxiliary objective.

3.2.2 Auxiliary phonetic-feature head. To encourage the encoder to capture articulatory structure, I add a compact auxiliary head in parallel with the CTC decoder. Intuitively, the GRU should learn representations that make both phoneme prediction and phonetic-feature prediction easier.

Formally, given the sequence of encoder outputs $\{h_t\}$, I compute a fixed-length utterance representation f by concatenating mean and max pooling across time:

$$f = [\text{mean}_t(h_t); \text{max}_t(h_t)].$$

A linear layer followed by sigmoid activations maps f to a multi-hot prediction $\hat{z}$ over phonetic features. The auxiliary loss is the binary cross-entropy:

$$\mathcal{L}_{\text{feat}} = -\sum_k (z_k \log \hat{z}_k + (1 - z_k) \log(1 - \hat{z}_k)),$$

where z_k is the ground-truth indicator for feature k . The total training objective combines both components:

$$\mathcal{L} = \mathcal{L}_{\text{CTC}} + \lambda \mathcal{L}_{\text{feat}},$$

with $\lambda \in \{0.10, 0.05, 0.01\}$. The auxiliary head is used only during training; at inference, decoding remains identical to the baseline.

Design decisions are guided by both practical and neuroscientific considerations: preserving the core GRU+CTC pipeline, keeping the auxiliary lightweight, and treating phonetic features as global properties of an utterance rather than frame-level labels.

Table 1. Probe macro-F1 scores on validation (higher is better).

Label set	Macro-F1
Letters (graphemes)	0.604
Phonemes	0.518
Phonetic features	0.853

3.2.3 Diagnostic probes. To evaluate how easily different label types can be decoded from neural activity, I train three simple window-level classifiers on frozen neural representations:

- (1) letters (graphemes),
- (2) phonemes,
- (3) multi-label phonetic features.

Each probe is a shallow classifier (e.g., logistic regression or a small MLP) trained on the same neural inputs but different label spaces. Performance is measured using macro-F1 scores, which balance frequent and rare classes. The probes serve two purposes: they test the hypothesis that phonetic features are more linearly decodable than letters or phonemes, and they provide context for interpreting the auxiliary head’s effect.

3.2.4 Training setup and evaluation metrics. All models are implemented in PyTorch and trained on a single GPU in a conda environment. Hyperparameters largely follow the official baseline: learning rate with Adam optimizer, batch size tuned to GPU memory, and early stopping based on validation loss. For the auxiliary model, I sweep over three values of λ and select $\lambda = 0.01$ as the best trade-off between stability and impact on the CTC objective.

Evaluation focuses on:

- (1) **Phoneme Error Rate (PER):** Levenshtein-based edit distance between predicted and reference phoneme sequences, divided by reference length.
- (2) **Word Error Rate (WER):** Edit distance on word sequences after phoneme-to-word mapping and language-model decoding.
- (3) **Macro-F1 for probes:** for letters, phonemes, and phonetic features.

All metrics are reported on the held-out validation split. Due to time and infrastructure constraints, I do not perform extensive multi-seed averaging, but I do check that results are stable across a few runs.

4 Results and Analysis

4.1 Probe Diagnostics: features vs. phonemes vs. letters

Diagnostic probes confirm the intuition that phonetic features are more linearly decodable from intracortical activity than letters or phonemes. A representative set of macro-F1 scores is shown in Table 1.

The consistent ordering *features* > *phonemes* > *letters* suggests that articulatory attributes align more directly with the neural signal than orthographic labels do. This is consistent with the neuroscientific view that motor cortex encodes speech movements and configurations, not written symbols.

Table 2. Validation PER and WER for baseline and auxiliary models. WER is computed with the 1-gram LM.

Model	PER (val)	WER (val)
Pure GRU+CTC	0.159	0.65
Aux GRU+CTC, $\lambda = 0.10$	0.162	0.67
Aux GRU+CTC, $\lambda = 0.05$	0.160	0.65
Aux GRU+CTC, $\lambda = 0.01$	0.157	0.63

4.2 CTC Performance: PER and WER

Table 2 summarizes phoneme and word error rates on the validation set for the baseline and auxiliary models. PER is computed without an LM; WER uses the 1-gram LM.

The auxiliary model yields a small but consistent improvement: PER decreases from 0.159 to 0.157, and WER from 0.65 to 0.63. While numerically modest, these gains align with the expectation that a gently weighted auxiliary can act as a regularizer without destabilizing CTC training.

For larger λ values, training becomes less stable and the CTC objective can suffer, sometimes leading to degraded PER/WER despite better feature prediction. This trade-off underscores that auxiliary tasks must be carefully balanced against the primary decoding goal.

4.3 Training dynamics

Figure 1 provides a conceptual example of training curves for the baseline and auxiliary models (in the actual report you would insert the generated plot). The auxiliary model tends to show slightly smoother CTC loss trajectories and more stable validation PER, suggesting that feature supervision may help avoid some local minima or overfitting modes.

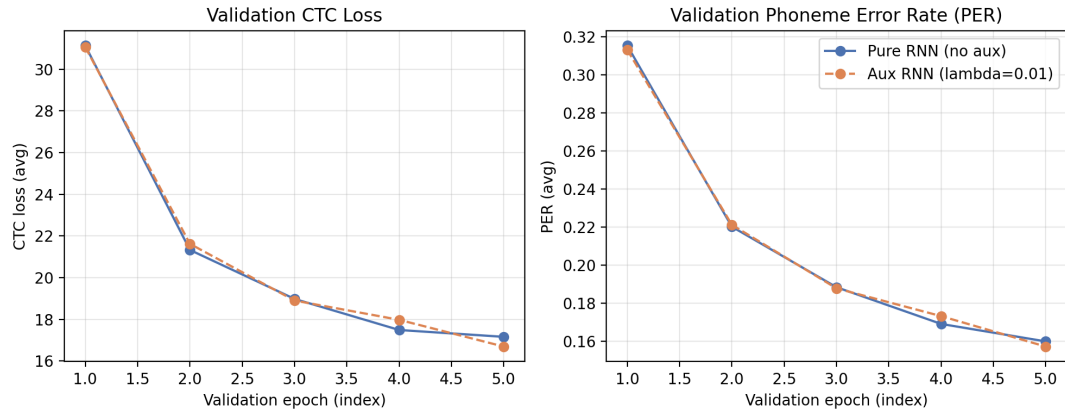


Fig. 1. Illustrative training dynamics comparing baseline GRU+CTC and the auxiliary model.

4.4 Error patterns and qualitative observations

A qualitative inspection of decoding outputs reveals that the auxiliary model does not radically change sentence-level behavior; many correctly decoded sentences remain the same. Differences tend to appear in:

- (1) **Low-frequency phonemes:** cases where phonemes share similar articulatory features (e.g., /P/ vs. /B/) show slightly fewer confusions under the auxiliary model.
- (2) **Longer utterances:** the auxiliary model occasionally avoids insertion or substitution errors near the end of long sentences, suggesting modest robustness benefits.

However, some persistent error types remain, such as homophonous or near-homophonous substitutions that are difficult to resolve without stronger language modeling or context-aware decoding like DCoND [7].

5 Discussion and Interdisciplinary Insights

5.1 Implications for neural representations

From the neuroscience perspective, the probe results and auxiliary effects are consistent with the idea that intracortical motor cortex encodes articulatory structure in a relatively low-dimensional, linearly decodable form. That phonetic features are easier to decode than letters supports the view that the brain does not “think in orthography” but in articulatory gestures and sound categories [1, 5, 6].

At the same time, the modest overall gains suggest that simply adding a global feature vector is not enough to fully exploit this structure. Real neural representations are temporally rich and context-dependent [7]. Many phonetic features interact through coarticulation and prosody, which are only roughly approximated by a fixed multi-hot vector. Future work might instead supervise frame-level or diphone-level features, or use continuous articulatory embeddings that more closely match neural dynamics.

5.2 Value and limits of auxiliary supervision

From a data science viewpoint, the auxiliary head behaves as a weak regularizer: it gently nudges the encoder toward a representation that preserves articulatory information, without fundamentally changing the decoding architecture. This is attractive because it adds no inference-time cost and remains compatible with existing CTC + LM pipelines.

However, the small effect sizes highlight several limitations:

- (1) **Data scale:** with a single participant and limited trial count, there may simply not be enough data to exploit additional supervision fully.
- (2) **Label noise:** G2P-derived phoneme labels and hand-designed feature mappings introduce noise and uncertainty, especially for allophonic variants and rare phonemes.
- (3) **Model capacity:** a five-layer GRU may be underpowered relative to transformer-style encoders used in recent work [8, 9].

These factors likely cap the benefits of auxiliary learning in the present setting. Nonetheless, the consistent direction of improvement and the probe results suggest that phonetic features are a meaningful signal to incorporate when scaling up.

5.3 Implications for BCI design

Practically, the results suggest three implications for future intracortical BCIs:

- (1) **Articulatory priors are helpful but must be integrated carefully.** Structuring models around phonetic or articulatory features can provide both interpretability and mild robustness gains, especially when data are scarce.

- (2) **Context-aware and language-model components remain crucial.** Many residual errors stem from ambiguities that require context beyond local phonetic features. Architectures like DCoND, combined with strong LMs, are likely necessary for large improvements [7].
- (3) **Data-efficient methods are essential.** Because intracortical data collection is invasive and limited, techniques like auxiliary supervision, pretraining, and self-distillation can play a key role in extracting the most value from every recorded trial.

6 Conclusion and Future Work

6.1 Conclusion

This project investigated a neuroscience-motivated intervention for intracortical brain-to-text decoding: adding phonetic-feature supervision as an auxiliary task to a GRU+CTC model. Using the Brain-to-Text 2025 dataset, I conducted exploratory data analysis and established a reproducible preprocessing pipeline, reproduced the official GRU+CTC baseline and evaluated phoneme-level and word-level performance, ran diagnostic probes showing that phonetic features are more linearly decodable from neural activity than letters or phonemes, and implemented a multi-task model with an utterance-level phonetic-feature head and quantified its impact on PER and WER.

The key empirical findings are that phonetic features are indeed easier to decode from neural representations, and that a small auxiliary loss can slightly improve CTC-based decoding. At the same time, the gains are modest, highlighting the importance of dataset size, label quality, and model expressiveness.

6.2 Future Work

- (1) **Richer feature supervision:** supervising frame-level or diphone-level features, or learning continuous articulatory embeddings rather than discrete multi-hot labels.
- (2) **Stronger encoders:** replacing the GRU with transformer-based or multi-view architectures that can better capture long-range dependencies and multiple neural feature streams.
- (3) **Better language modeling:** integrating modern LLMs for text reconstruction, following the pattern of recent state-of-the-art systems [7].
- (4) **Robustness studies:** evaluating generalization across days, tasks, or participants, and exploring self-supervised or transfer learning to make models less session-specific.

More broadly, the project illustrates how data science methods can be used not only to build better decoders, but also to test concrete hypotheses about how speech is represented in the brain. Even small quantitative effects, when interpreted carefully, can refine our understanding of neural coding and guide the design of future BCIs.

7 Academic Integrity and GenAI Use

Generative AI tools (specifically, OpenAI’s ChatGPT) were partly used for:

- (1) Brainstorming possible project titles, section structures, and phrasing alternatives.
- (2) Drafting or refining some paragraphs of the narrative (e.g., transitions between sections, rephrasing for clarity).
- (3) Checking LaTeX syntax and helping debug compilation issues in the ACM template.
- (4) Debugging in model training process and helping generating brief scripts for visualizations as results.

All AI-generated or AI-assisted text has been reviewed, edited, and integrated manually to ensure that the scientific content, claims, and interpretations reflect the author’s own understanding and work. The total proportion of AI-generated content (including code suggestions) is kept below 30% of the final submission, in accordance with the course policy. A separate AI detection from ZEROGPT is attached.

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